SMARTBRIDGE
Let's Bridge the Gap

Project Initialization and Planning Phase

Date	15 March 2024
Team ID	EN25CS3090269
Project Title	Smart Healthcare Analytics
Maximum Marks	3 Marks

Project Proposal (Proposed Solution) template

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

Project Overview	
Objective	To develop a Smart Healthcare Analytics system that transforms synthetic patient data into meaningful visualisations and insights using Tableau. The project also aims to provide a Flask-based website for presenting the project and accessing its dashboard and Story.
Scope	The project focuses on analysing patient demographics, BMI, blood pressure, exercise levels, alcohol and smoking consumption, and recorded disease indicators such as Diabetes, Heart Disease, and Stroke. It includes dataset preparation, interactive Tableau dashboards, a Tableau Story, and a Flask website. The project is intended for educational and exploratory analysis, not clinical diagnosis.
Problem Statement	
Description	Healthcare datasets contain large amounts of patient information, which can be difficult to understand when presented only in raw tabular format. Analysing patient demographics, lifestyle factors, and health indicators manually can make it difficult to identify patterns and compare different categories. Therefore, a visual analytics solution is needed to present healthcare information in a clear and organised manner.
Impact	Without proper visualisation, understanding patient distributions and comparing health indicators can be time-consuming. A centralised interactive dashboard can make data exploration easier, improve the presentation of information, and help users identify patterns within the dataset.

Proposed Solution	
Approach	The project uses a healthcare dataset containing 5,000 patient records and 21 fields. The data is connected to Tableau Desktop, where calculated fields, aggregations, charts, KPI cards, heatmaps, and interactive filters are used to analyse and present healthcare information. Tableau Public is used to publish the dashboard and Story. A Flask website developed using Python, HTML, and CSS provides project information and access to the visualisations.
Key Features	1. Interactive healthcare analytics dashboard. 2. KPI cards for Total Patients, Diabetes, Heart Disease, and Stroke. 3. Patient distribution analysis by age group and gender. 4. Average blood pressure comparison across age groups. 5. BMI and heart disease analysis using charts and heatmaps. 6. Exercise level distribution visualisation. 7. Filters for Age Group, Gender, and BMI Category. 8. Tableau Story for presenting healthcare observations. 9. Flask-based website with Home, About, Dashboard, Story, and Contact sections. 10. GitHub repository for source code and project documentation.

Resource Requirements

Resource Type	**Description**	**Specification/Allocation**
Hardware		
Computing Resources	Computer or laptop for data analysis, Tableau development, and Flask website development.	Standard personal computer/laptop with a modern multi-core processor. No dedicated GPU required.
Memory	RAM required to run Tableau Desktop, Python, Flask, and a web browser.	8 GB RAM recommended.

Storage	Storage for the dataset, Tableau workbook, website files, and project documentation.	Available local disk space; approximately 1 GB or more is a practical allocation.
Software		
Frameworks	Framework used for developing the website.	Python Flask
Libraries	Software libraries used for website development, if required.	Flask and its required dependencies.
Development Environment	Used to create healthcare charts, dashboards, and Stories.	Tableau Desktop and Tableau Public
Frontend Technologies	Used to design and develop the website interface.	HTML and CSS
Development Environment	Used to write, run, and manage project files.	Python environment, Visual Studio Code or another code editor, and web browser.
Version Control	Used to store and manage source code and project documentation.	Git and GitHub
Data		
Data	Healthcare dataset used for analysis and visualisation.	smart_healthcare_dataset.csv
Data Attributes	Patient demographics, lifestyle information, and health indicators.	Age, Age Group, Gender, BMI, BMI Category, Blood Pressure, Exercise Level, Alcohol, Smoking, Diabetes, Heart Disease, and Stroke.
Data Usage	Used to create calculated fields, analyse health indicators, and develop interactive visualisations.	Tableau Desktop